

someone up to speed on a subject without having to write up a summary e-mail but, if proper care is not exercised, forwarding e-mails can create a significant security threat. As an e-mail is forwarded, the recipients of the mail (until that point in time) are automatically listed in the body of the e-mail.

As the chain keeps moving forward, more and more recipient IDs are placed on the list. Unfortunately, if a spammer or someone just looking to make a quick buck gets hold of the e-mail, he can sell the entire list of e-mail IDs and then everyone could start receiving spam. It only takes a few seconds to delete all the previous recipient IDs before forwarding a piece of mail. You can this avoid the terrible situation of you being the cause of all your friends or coworkers getting spammed.

Making backups and keeping records

1. Failing to back up e-mails. Many a times e-mails are used to make legally binding contracts, major financial decisions and conduct professional meetings. Just as we keep a hard copy of other important business and personal documents, it is important to regularly back up these important e-mails to preserve a record. This will be helpful in the scenario when an e-mail client crashes and entire data is lost. The frequency of backups depends on e-mail usage, but under no circumstances should it be done less frequently than every three months.

2. Mobile access. Presuming a backup exists. Mobile e-mail access, such as through Android/smart phones/BlackBerry, has revolutionised the way we think about e-mail; no longer it is tied to a PC, but rather it can be checked on-the-go anywhere. Many a times, BlackBerry users simply assume that a copy of the e-mails they check and delete off the BlackBerry will still be available on their home or office computer.

But it is important to keep in mind that some e-mail servers and

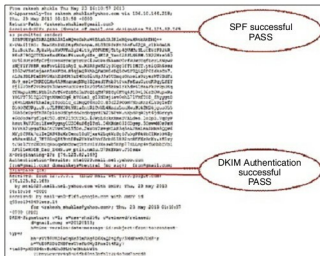


Fig. 10: DKIM/SPF authentication and validation successful

client software download e-mails to the BlackBerry device and then delete them from the server. Thus, for some mobile e-mail access devices, if e-mail is deleted from the device, it is deleted from the Inbox. Just be aware of the default settings of e-mail client and ensure to keep a copy of the retained e-mail. It also happens in the case of MS Outlook that the e-mail is downloaded onto the PC. Here I would like to mention that it is the protocol which does it. By protocol I mean POP3, which downloads all the e-mails onto the hard disk and clears them from the e-mail server until explicitly told by the setting. This setting is shown in Fig. 11. By default, this setting is unchecked in MS Outlook, so all the e-mails when downloaded on the local hard disk get deleted from the e-mail server.

3. Thinking that an erased e-mail is gone forever. It is to be noted that even after deleting an e-mail message from Inbox and the Send folder, it often exists in backup folders on remote servers for years, and can be retrieved by skilled professionals. So e-mail can be like a permanent document.

Avoiding fraudulent e-mail

1. Prize/lottery/scam mails. Spammers use a wide variety of clever titles, which often include social engineering to get one to open e-mails which they fill with all sorts of bad things, such as:

- (i) Winning of the Irish lotto,

the Yahoo lottery, or any other big cash prize

(ii) Nigerian king or prince trying to send \$10 million

(iii) Bank account details reconfirmation immediately. This is a common phishing attack

(iv) Unclaimed inheritance

(v) Resending the mail not sent as 'Returned Mail'

(vi) The news headline e-mail

(vii) Winning an iPod Nano e-mail

2. Not recognising phishing attacks in e-mail content. While never opening a phishing e-mail is the best way to secure your computer, even the most experienced e-mail user will occasionally accidentally open up a phishing e-mail. At this point, the key to limiting your damage is recognising the phishing e-mail for what it is. Phishing is a type of online fraud wherein the sender of the e-mail tries to trick you into giving out personal passwords or banking information. The sender will typically steal the logo from a well-known bank or PayPal and try to format the e-mail to look like it came from the bank.

Usually, the phishing e-mail asks to click on a link in order to confirm banking information or password, but it may just ask to reply to the e-mail with personal information. Whatever form the phishing attempt takes, the goal is to fool you into entering your information into something which appears to be safe and secure, but in fact it is just a dummy site set up by the scammer. If you provide the phisher with personal information, the information will help the scammer to steal identity and money from your accounts.

3. Signs of phishing. You can identify a phishing e-mail from:

(i) A logo that looks distorted or stretched

(ii) E-mail that refers to as 'Dear Customer' or 'Dear User' rather than